

2×2 mechanism decomposition: substrate × baseline (forecaster task, n=9 per cell)

baseline reference

**LTC-liquid
reference**

**nonliquid
reference**

**classical MLP
substrate**

liquid_via_classical

τ -isolating (classical substrate)

$$\Delta = +0.115$$

quantum_via_nonliquid

control

$$\Delta = -0.167$$

cell substrate

**quantum cell
substrate**

liquid_via_quantum

τ -isolating (quantum substrate)

$$\Delta = -0.334$$

quantum_via_ltc

control

$$\Delta = -0.616$$

$$\Delta_{\text{combined}} = -0.501 \quad (95\% \text{ CI excludes 0 negatively}; \quad |\Delta_{\tau, \text{classical}} - \Delta_{\tau, \text{quantum}}| = 0.449)$$